Unit 1: Unit I: Matrices (10 Hours)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 35 mins

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1. Rank of a Matrix & Normal (Canonical) Form

1. In-Depth Mathematical Foundations of Matrix Rank

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous Linear Algebra Definition: Let A be an $m \times n$ matrix with entries in $\mathbb{R}$ (or $\mathbb{C}$). The **Rank** of A, denoted by $\rho(A)$ or $\text{rank}(A)$, is defined as the dimension of the vector space spanned by its row vectors (Row Rank) or column vectors (Column Rank). By the Fundamental Rank Theorem of Linear Algebra:

$$\text{Row Rank}(\mathbf{A}) = \text{Column Rank}(\mathbf{A}) = \rho(\mathbf{A}) \leq \min(\mathbf{m}, \mathbf{n})$$

Minor-Based Criterion of Rank:

A matrix A is said to be of rank r if and only if:

1. There exists **at least one minor** (determinant of a square submatrix) of order r that is strictly non-zero: $|M_r| \neq 0$.
2. **Every minor** of order $(r+1)$ and higher vanishes identically: $|M_{r+1}| = 0$.

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2. Deep-Dive: Row Echelon Form Reduction

Strict Mathematical Criteria for Row Echelon Form:

A matrix E is in **Row Echelon Form** if:

1. **Zero Rows at Bottom:** All rows consisting entirely of zeros are grouped at the very bottom.
2. **Leading Entry (Pivot):** The leading coefficient (the first non-zero entry from the left) of each non-zero row is strictly to the right of the leading coefficient of the row directly above it.
3. **Zeros Below Pivots:** All entries in a column directly below a pivot entry are zero.

$$\rho(\mathbf{A}) = \text{Total Number of Non-Zero Rows in Row Echelon Form}$$

Invariance of Rank under Elementary Row Transformations:

Elementary row transformations do not alter the linear span of the row vectors:

1. Interchanging two rows: $R_i \leftrightarrow R_j$ (Permutation matrix P, $|P| = -1 \neq 0$).

2. Multiplying a row by a non-zero scalar: $R_i \to kR_i,\ k \neq 0$ (Diagonal scaling matrix D, $|D| = k \neq 0$).

3. Adding a scalar multiple of one row to another: $R_i \to R_i + cR_j$ (Shear matrix S, $|S| = 1 \neq 0$).

Because all elementary matrices are non-singular ($|E| \neq 0$), rank is strictly invariant: $\rho(EA) = \rho(A)$.

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3. Deep-Dive: Normal (Canonical) Form $[I_r\ 0; 0\ 0]$

By using **both elementary row AND elementary column operations**, any arbitrary $m \times n$ matrix A of rank r can be brought into its unique Canonical Normal Form:

$$\mathbf{N} = \begin{bmatrix} I_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix}$$

Existence of Non-Singular Matrices P and Q:

There exist non-singular square matrices P (of order m) and Q (of order n) representing the product of all row operations and column operations respectively, such that:

$$\mathbf{P} \cdot \mathbf{A} \cdot \mathbf{Q} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

- **Procedure to compute P and Q simultaneously:**

1. Write $A_{m\times n} = I_m \cdot A \cdot I_n$.
2. Perform elementary row operations on A and simultaneously on the pre-factor I_m.
3. Perform elementary column operations on A and simultaneously on the post-factor I_n.
4. When the central matrix reduces to $[I_r\ 0; 0\ 0]$, the left matrix becomes P and the right matrix becomes Q!

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4. Fundamental Properties & Sylvester's Rank Inequalities

1. **Sub-multiplicative Property:** $\rho(AB) \leq \min(\rho(A), \rho(B))$.

2. **Sylvester's Law of Nullity:** For an $m \times n$ matrix A and $n \times p$ matrix B:

$$\rho(\mathbf{A}) + \rho(\mathbf{B}) - \mathbf{n} \leq \rho(\mathbf{AB}) \leq \min(\rho(\mathbf{A}), \rho(\mathbf{B}))$$

3. **Rank of Transpose & Conjugate:** $\rho(A^T) = \rho(A)$ and $\rho(A^T A) = \rho(AA^T) = \rho(A)$.

4. **Rank of Sum:** $\rho(A + B) \leq \rho(A) + \rho(B)$.

Problem:

Find the rank of matrix $A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$.

Step 1 🚀: Compute the Determinant of A

$$|A| = (2)(6) - (4)(3) = 12 - 12 = 0$$

Since $|A| = 0$, the rank cannot be 2: $\rho(A) < 2$.

Step 2 ⚡: Check Order-1 Minors (Single Entries)

The matrix contains non-zero entries (e.g., $a_{11} = 2 \neq 0$). Therefore, there is at least one non-vanishing minor of order 1.

Step 3 🎯: Row Echelon Verification ($R_2 \to R_2 - 1.5R_1$)

$$\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}$$

There is exactly **1 non-zero row**.

🎯 Rank of A: $\rho(\mathbf{A}) = \mathbf{1}$.

University Question (4 Marks):

Reduce the matrix A to Row Echelon form and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

Step 1 🚀: Eliminate entries below pivot in Column 1

Apply row operations $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Eliminate entry below pivot in Column 3

Apply $R_3 \to R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3 🎯: Count Non-Zero Rows

The echelon form contains 2 non-zero rows ($[1, 2, 3]$ and $[0, 0, 1]$) and 1 zero row.

🎯 Row Echelon Rank: $\rho(\mathbf{A}) = \mathbf{2}$.

University Question (6 Marks - MMDU Annual Exam):

Reduce matrix A to its Normal Form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ using elementary row AND column transformations, and find non-singular matrices P and Q such that $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 1 🚀: Setup Identity Matrices $A = I_3 A I_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Elementary Row Operation $R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A I_3$$

Step 3 🎯: Elementary Column Operations $C_2 \to C_2 - C_1,\ C_3 \to C_3 - 2C_1$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\\0 & 1 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\0 & 0 & 1\end{bmatrix} A \begin{bmatrix}1 & -1 & -2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

Step 4 🏁: Row Operation $R_3 \to R_3 - R_2$ and Column Operation $C_3 \to C_3 - C_2$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\1 & -1 & 1\end{bmatrix} A \begin{bmatrix}1 & -1 & -1\\0 & 1 & -1\\0 & 0 & 1\end{bmatrix}$$

This is the standard Normal Form $\begin{bmatrix}I_2 & 0\\0 & 0\end{bmatrix}$.

🎯 Canonical Normal Form: $\begin{bmatrix}I_2 & 0\\0 & 0\end{bmatrix}$ | Rank $\rho(\mathbf{A}) = \mathbf{2}$ | $P = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\1 & -1 & 1\end{bmatrix}$, $Q = \begin{bmatrix}1 & -1 & -1\\0 & 1 & -1\\0 & 0 & 1\end{bmatrix}$.

2. Gauss-Jordan Method for Matrix Inverse [A | I] ~ [I | A^{-1}]

1. The Gauss-Jordan Inversion Algorithm (Theoretical Analysis)

📌 **CRITICAL EXAM DERIVATION:**
University Exam Distinction: Never use the classical Adjugate formula $A^{-1} = \frac{\text{adj}(A)}{|A|}$ when the question explicitly states **"Find inverse using Gauss-Jordan method"**! The classical method requires $O(n!)$ operations via minors, whereas Gauss-Jordan requires only $O(n^3)$ row operations.

Mathematical Principle of Gauss-Jordan Inversion:

Let A be an $n \times n$ invertible matrix ($|A| \neq 0$). If a sequence of elementary row operations $E_1, E_2, \dots, E_k$ transforms A into the identity matrix I:

$$(E_k \dots E_2 E_1)A = I \implies \mathbf{A^{-1} = E_k \dots E_2 E_1}$$

Therefore, if we apply the identical sequence of operations to the augmented matrix $[A \mid I_n]$:

$$(E_k \dots E_2 E_1)[A \mid I_n] = [(E_k \dots E_1)A \mid (E_k \dots E_1)I_n] = [\mathbf{I_n \mid A^{-1}}]$$

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2. Step-by-Step Algorithmic Execution Pipeline

1. **Augmentation:** Construct the partitioned block matrix $[A \mid I_n]$ of dimension $n \times 2n$.

2. **Forward Elimination (Upper Triangularization):**

- For column $j = 1, 2, \ldots, n$:
- **Pivoting:** Find row $i \geq j$ with a non-zero pivot a_{ij} (preferring 1 or -1). Swap $R_j \leftrightarrow R_i$.
- If pivot entry $a_{jj} = 0$, matrix is **singular** (A^{-1} does not exist!).
- Normalize the pivot row: $R_j \to \frac{1}{a_{jj}} R_j$.
- Eliminate all entries below pivot: for each $k > j$, $R_k \to R_k - a_{kj} R_j$.

3. **Backward Elimination (Diagonalization):**

- Starting from the bottom row $j = n, n-1, \ldots, 2$:
- For each row $k < j$, eliminate the above-pivot entries: $R_k \to R_k - a_{kj} R_j$.

4. **Extraction:** The right half of the augmented matrix is strictly A^{-1}. Verify that $A \cdot A^{-1} = I_n$.

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3. Comparison of Matrix Inversion Techniques

Feature	Gauss-Jordan Method	Adjugate Formula	LU Decomposition
Operation Count	n^3 additions & multiplications	$\sim (n+1)!$ determinant operations	$\frac{2}{3}n^3$ operations
Memory Required	$n \times 2n$ augmented grid	High cofactor matrix cache	$n \times n$ in-place factorization
Suitability	Hand calculation in university exams	2×2 matrices only	High-performance computer algorithms

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Problem:

Find the inverse of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ using the Gauss-Jordan method.

Step 1 🚀: Setup Augmented Matrix $[A \mid I]$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Make entry below pivot zero ($R_2 \to R_2 - 3R_1$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

Step 3 🎯: Normalize pivot of row 2 ($R_2 \to -\frac{1}{2}R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

Step 4 🏁: Eliminate entry above pivot ($R_1 \to R_1 - 2R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

🎯 Final Inverse: $\mathbf{A^{-1}} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$. Verified: $A \cdot A^{-1} = I$.

University Question (6 Marks - MMDU Question Paper):
Employ the Gauss-Jordan method to determine the inverse of:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Step 1 🚀: Swap Rows to Get Leading Pivot 1 ($R_1 \leftrightarrow R_2$)

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Eliminate Column 1 ($R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right]$$

Step 3 🎯: Swap $R_2 \leftrightarrow R_3$, Normalize and Eliminate

$$
\begin{aligned}
R_2 \to -R_3: &\quad [0,1,-1 \mid 0,1,-1] \\
R_3 \to R_3 + 3R_2: &\quad [0,0,-4 \mid 1,1,-3] \\
R_3 \to -\frac{1}{4}R_3: &\quad [0,0,1 \mid -\frac{1}{4},-\frac{1}{4},\frac{3}{4}]
\end{aligned}
$$

Step 4 🏁: Eliminate Above Pivots to reach $[I \mid A^{-1}]$

$$
\begin{aligned}
R_2 \to R_2 + R_3: &\quad [0,1,0 \mid -\frac{1}{4},\frac{3}{4},-\frac{1}{4}] \\
R_1 \to R_1 - R_3 - 2R_2: &\quad [1,0,0 \mid \frac{3}{4},-\frac{1}{4},-\frac{1}{4}]
\end{aligned}
$$

🎯 Complete Inverse: $\mathbf{A^{-1}} = \frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$. Verified: $|A| = 4 \neq 0$.

3. Consistency of System of Linear Equations AX = B

1. Comprehensive Theory: System of Linear Equations $AX = B$

A system of m linear equations in n unknowns $x_1, x_2, \ldots, x_n$ is represented in matrix notation as:

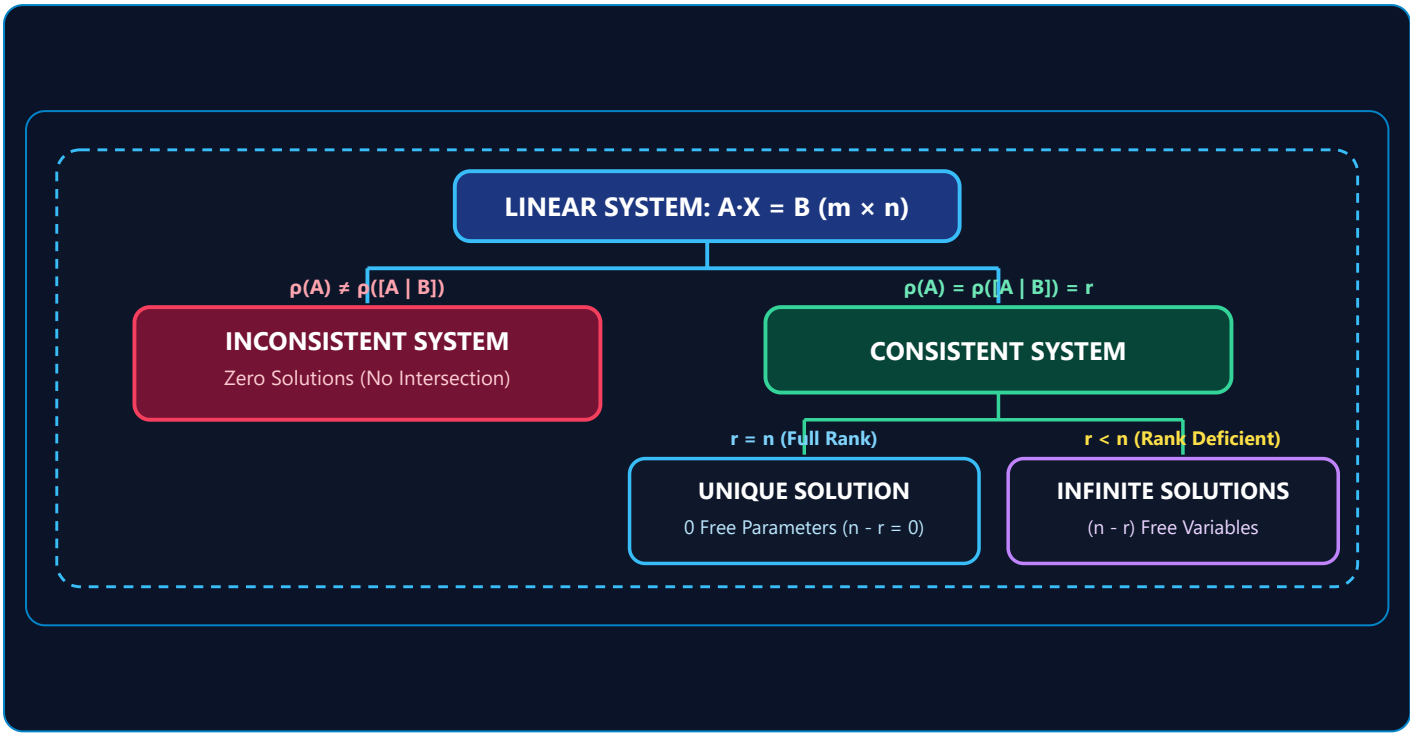
$$
\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \iff \mathbf{AX = B}
$$

The Rouché–Capelli Theorem:

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: A system of linear equations $AX = B$ is **consistent** (possesses at least one solution) if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $[A \mid B]$:

$$
\rho(\mathbf{A}) = \rho([\mathbf{A} \mid \mathbf{B}])
$$

2. Complete Classification of Solution Regimes



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3. Homogeneous Linear Systems $AX = 0$

For $AX = 0$, $B = 0$. Since augmenting a column of zeros cannot increase matrix rank, $\rho(A) = \rho([A \mid 0])$ always holds. Therefore, **a homogeneous system is ALWAYS consistent** and always possesses at least the trivial solution $X = 0$.

Nature of Solutions for Square Systems ($n \times n$):

1. **Trivial Solution ($X = 0$):**

- Occurs when $\rho(A) = n \iff |A| \neq 0$ (Non-singular matrix).
- The columns of A are linearly independent; the only solution is $x_1 = x_2 = \cdots = x_n = 0$.

2. **Non-Trivial Solutions (Infinite Non-Zero Solutions):**

- Occurs when $\rho(A) = r < n \iff |A| = 0$ (Singular matrix), yielding $(n - r)$ linearly independent parameters.

> 🚨 **EXAMINER TRAP / PITFALL:**
> **Examiner's Trap:** When an exam problem asks: "*Find values of k for which the homogeneous equations have non-trivial (non-zero) solutions*", simply set the determinant to zero: $\mathbf{|A| = 0}$!

University Question (6 Marks - Classic MMDU Exam Problem):
Investigate for what values of λ and μ the system of equations:

$$\begin{cases} x+y+z=6 \\ x+2y+3z=10 \\ x+2y+\lambda z=\mu \end{cases}$$

has: (i) No solution, (ii) A unique solution, (iii) Infinitely many solutions.

Step 1 🚀: Setup Augmented Matrix and Echelon Reduction

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

Step 2 ⚡: Case Analysis by Rouché–Capelli Theorem

1. Case (i) No Solution (Inconsistent): $\lambda = 3,\ \mu \neq 10 \implies \rho(A) = 2 \neq \rho([A \mid B]) = 3$
2. Case (ii) Unique Solution: $\lambda \neq 3,\ \forall \mu \implies \rho(A) = \rho([A \mid B]) = 3 = n$
3. Case (iii) Infinite Solutions: $\lambda = 3,\ \mu = 10 \implies \rho(A) = \rho([A \mid B]) = 2 < n\ (z = k)$

🎯 **Verified Result:** (i) No solution for $\lambda = 3, \mu \neq 10$ | (ii) Unique for $\lambda \neq 3$ | (iii) Infinite for $\lambda = 3, \mu = 10$.

4. Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem

1. Eigenvalues (Characteristic Roots) & Eigenvectors

✅ **FUNDAMENTAL LAW / RULE:**

Definition: Let A be an $n \times n$ square matrix. A scalar λ is called an **Eigenvalue** (or characteristic root / latent root) of A if there exists a **non-zero** column vector $X \neq 0$ such that:

$$\mathbf{AX} = \lambda \mathbf{X} \iff (\mathbf{A} - \lambda \mathbf{I})\mathbf{X} = \mathbf{0}$$

> The non-zero vector X is called the **Eigenvector** (or characteristic vector) corresponding to λ.

Characteristic Equation & Matrix Invariants (3×3 Shortcut):

For $AX = \lambda X$ to have a non-trivial solution $X \neq 0$, the matrix $(A - \lambda I)$ must be singular:

$$\mathbf{|A - \lambda I| = 0}$$

For a 3×3 matrix A, the characteristic polynomial expands directly using trace and principal minors:

$$\mathbf{\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0}$$

Where:

- $S_1 = \text{Tr}(A) = a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3$ *(Sum of main diagonal elements = Sum of eigenvalues)*
- $S_2 = M_{11} + M_{22} + M_{33}$ *(Sum of cofactors/minors of main diagonal elements)*
- $|A| = \det(A) = \lambda_1 \cdot \lambda_2 \cdot \lambda_3$ *(Determinant = Product of eigenvalues)*

Key Properties of Eigenvalues:

1. The eigenvalues of an **upper or lower triangular (or diagonal) matrix** are simply its main diagonal elements.
2. The eigenvalues of a **real symmetric matrix** ($A^T = A$) are always **purely real**, and eigenvectors of distinct eigenvalues are mutually **orthogonal** ($X_i^T X_j = 0$).
3. If λ is an eigenvalue of A, then:

- λ^k is an eigenvalue of A^k.
- $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} (for $|A| \neq 0$).
- $\frac{|A|}{\lambda}$ is an eigenvalue of $\text{adj}(A)$.

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2. Cayley-Hamilton Theorem & Its Applications

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Theorem Statement: "Every square matrix satisfies its own characteristic equation."**

> If the characteristic equation of an $n \times n$ matrix A is:

$$|A - \lambda I| = (-1)^n (\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \cdots + c_n) = 0$$

> Then replacing scalar λ by matrix A and constant c_n by $c_n I$ yields the zero matrix:

$$\mathbf{A^n + c_1A^{n-1} + c_2A^{n-2} + \cdots + c_nI = 0}$$

Two Primary Applications of Cayley-Hamilton Theorem:

1. **Finding Matrix Inverse A^{-1} (when $c_n = (-1)^n|A| \neq 0$):**

Multiply the matrix equation by A^{-1} and isolate A^{-1}:

$$\mathbf{A^{-1} = -\frac{1}{c_n}(A^{n-1} + c_1A^{n-2} + \cdots + c_{n-1}I)}$$

2. **Computing Higher Matrix Powers (A^4A^8, etc.):**

Express A^n in terms of lower powers $(A^{n-1}, \ldots, A, I)$ and substitute iteratively.

University Question (6 Marks - MMDU Annual Exam):
For matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$: (1) Verify Cayley-Hamilton Theorem, (2) Evaluate A^{-1}, and (3) Compute A^4.

Step 1 🚀: Characteristic Equation via Invariants $S_1, S_2, |A|$

- $S_1 = \text{Tr}(A) = 1 + 1 + 1 = 3$

- $S_2 = M_{11} + M_{22} + M_{33} = (1-1) + (1-3) + (1-0) = 0 - 2 + 1 = -1$

- $|A| = 1(0) - 2(3) + 1(-3) = -9$

- Characteristic equation: $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0$.

Step 2 ⚡ : Verify $A^3 - 3A^2 - A + 9I = 0$

$$A^2 = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix}, \quad A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 11 & 1 \\ -9 & -2 & -7 \\ 21 & 11 & 7 \end{bmatrix}$$

Substituting into $A^3 - 3A^2 - A + 9I$:

$$\begin{bmatrix} 4-12-1+9 & 11-9-2+0 & 1-0-1+0 \\ -9+9-0+0 & -2-6-1+9 & -7+6+1+0 \\ 21-18-3+0 & 11-12+1+0 & 7-15-1+9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(Verified!)}$$

Step 3 🎯 : Deduce A^{-1} and A^4

- Multiply by A^{-1}: $A^2 - 3A - I + 9A^{-1} = 0 \implies$

$$A^{-1} = -\frac{1}{9}(A^2 - 3A - I) = \frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -1 \\ 3 & -7 & -1 \end{bmatrix}$$

- Multiply by A: $A^4 = 3A^3 + A^2 - 9A$.

🎯 **Verified Result:** Cayley-Hamilton verified ($A^3 - 3A^2 - A + 9I = 0$) | $A^{-1} = -\frac{1}{9}(A^2 - 3A - I)$.

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Mathematics I (`BMAT-001`)

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 1 (1 Mark): Rank of a Matrix & Row Echelon Form

Question:
Define the rank of a matrix and state the rank of a matrix in Row Echelon Form.

Step-by-Step Solution:
The rank $\rho(A)$ of an $m \times n$ matrix A is the order of its largest non-vanishing minor (or the number of linearly independent rows). In Row Echelon Form, $\rho(A)$ equals the **total number of non-zero rows**.

🎯 **Verified Result:** $\rho(A)$ = Number of non-zero rows in echelon form.

📘 Question 2 (1 Mark): Eigenvalues of Upper Triangular Matrix

Question:
Find the sum and product of the eigenvalues of $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Step-by-Step Solution:
Since A is upper triangular, its eigenvalues are its diagonal elements: $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = 5$.

- **Sum of eigenvalues** = $\text{Tr}(A) = 3 + 2 + 5 = 10$.
- **Product of eigenvalues** = $|A| = 3 \times 2 \times 5 = 30$.

🎯 **Verified Result:** Sum = 10, Product = 30.

📘 Question 13 (2 Marks): Inverse via Cayley-Hamilton Theorem

Question:
State Cayley-Hamilton Theorem. If $A^2 - 4A + 3I = 0$, find A^{-1} in terms of A and I.

Step-by-Step Solution:
Every square matrix satisfies its own characteristic equation $|A - \lambda I| = 0$.
Pre-multiplying $A^2 - 4A + 3I = 0$ by A^{-1}:

$$A - 4I + 3A^{-1} = 0 \implies A^{-1} = \frac{1}{3}(4I - A)$$

🎯 **Verified Result:** $A^{-1} = \frac{1}{3}(4I - A)$.